

# Double Perovskite Curated Dataset

Supplementary Material: Curated database of double perovskite compounds and DFT-calculated properties used for training models.

## Sheet 1: True Double Perovskites (N = 416 samples)

| Material ID | Formula     | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|-------------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-18967    | Ba2CoMoO6   | -2.4633                    | 0.0000                 | 0.0000        | Fm-3m       | Ba       | Co       | Mo       |
| mp-561894   | Ba2CoMoO6   | -2.4541                    | 0.0092                 | 1.2740        | I4/mmm      | Ba       | Co       | Mo       |
| mp-545792   | Ba2CoReO6   | -2.4479                    | 0.0487                 | 0.1982        | R-3m        | Ba       | Re       | Co       |
| mp-1228653  | Ba2CoRuO6   | -2.0998                    | 0.0192                 | 0.0000        | P3m1        | Ba       | Co       | Ru       |
| mp-1274542  | Ba2CoWO6    | -2.5654                    | 0.0108                 | 1.8427        | I4/mmm      | Ba       | Co       | W        |
| mp-22112    | Ba2CoWO6    | -2.5762                    | 0.0000                 | 0.0000        | R-3m        | Ba       | Co       | W        |
| mp-1079901  | Ba2CrMoO6   | -2.6602                    | 0.0000                 | 0.0000        | I4/mmm      | Ba       | Cr       | Mo       |
| mp-1277787  | Ba2FeMoO6   | -2.5003                    | 0.0008                 | 0.8641        | C2/m        | Ba       | Fe       | Mo       |
| mp-1288525  | Ba2FeMoO6   | -2.5010                    | 0.0001                 | 0.7715        | C2          | Ba       | Fe       | Mo       |
| mp-1288978  | Ba2FeMoO6   | -2.5011                    | 0.0000                 | 0.9037        | C2          | Ba       | Fe       | Mo       |
| mp-18995    | Ba2FeMoO6   | -2.4978                    | 0.0033                 | 0.0000        | C2/m        | Ba       | Fe       | Mo       |
| mp-31756    | Ba2FeReO6   | -2.4859                    | 0.0030                 | 0.0000        | Fm-3m       | Ba       | Re       | Fe       |
| mp-1272506  | Ba2FeWO6    | -2.6398                    | 0.0003                 | 2.5062        | P-1         | Ba       | Fe       | W        |
| mp-18845    | Ba2FeWO6    | -2.6402                    | 0.0000                 | 0.0000        | I4/mmm      | Ba       | Fe       | W        |
| mp-2217283  | Ba2MgFeMoO6 | -2.3811                    | 0.1941                 | 1.5819        | P1          | Ba       | Fe       | Mo       |
| mp-20399    | Ba2MnMoO6   | -2.6297                    | 0.0000                 | 1.2654        | Fm-3m       | Ba       | Mn       | Mo       |
| mp-1228414  | Ba2MnNbO6   | -2.9407                    | 0.0274                 | 0.0000        | Fm-3m       | Ba       | Nb       | Mn       |
| mp-1272179  | Ba2MnNbO6   | -2.9635                    | 0.0046                 | 1.1628        | Pnma        | Ba       | Nb       | Mn       |
| mp-1275240  | Ba2MnNbO6   | -2.9629                    | 0.0052                 | 1.1475        | P-1         | Ba       | Nb       | Mn       |
| mp-1278519  | Ba2MnNbO6   | -2.9516                    | 0.0165                 | 0.0764        | Imma        | Ba       | Nb       | Mn       |
| mp-1278534  | Ba2MnNbO6   | -2.9617                    | 0.0064                 | 0.1858        | I4/mmm      | Ba       | Nb       | Mn       |
| mp-20813    | Ba2MnReO6   | -2.6379                    | 0.0000                 | 0.0000        | Fm-3m       | Ba       | Re       | Mn       |
| mp-1228678  | Ba2MnRuO6   | -2.3171                    | 0.0229                 | 0.0000        | C2/m        | Ba       | Mn       | Ru       |

| Material ID | Formula                            | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group          | element1 | element3 | element4 |
|-------------|------------------------------------|----------------------------|------------------------|---------------|----------------------|----------|----------|----------|
| mp-19085    | Ba <sub>2</sub> MnWO <sub>6</sub>  | -2.7470                    | 0.0000                 | 2.2267        | Fm-3m                | Ba       | Mn       | W        |
| mp-1228382  | Ba <sub>2</sub> NbCoO <sub>6</sub> | -2.7107                    | 0.0661                 | 1.5346        | Fm-3m                | Ba       | Co       | Nb       |
| mp-31627    | Ba <sub>2</sub> NbCrO <sub>6</sub> | -3.0562                    | 0.0000                 | 1.5824        | P6 <sub>3</sub> /mmc | Ba       | Cr       | Nb       |
| mp-1021504  | Ba <sub>2</sub> NbFeO <sub>6</sub> | -2.8956                    | 0.0000                 | 1.4550        | Fm-3m                | Ba       | Fe       | Nb       |
| mp-1022733  | Ba <sub>2</sub> NbFeO <sub>6</sub> | -2.8774                    | 0.0181                 | 1.6908        | P-3m1                | Ba       | Fe       | Nb       |
| mp-1022808  | Ba <sub>2</sub> NbFeO <sub>6</sub> | -2.8733                    | 0.0223                 | 1.3498        | P6 <sub>3</sub> /mmc | Ba       | Fe       | Nb       |
| mp-1205833  | Ba <sub>2</sub> NbRhO <sub>6</sub> | -2.7332                    | 0.0000                 | 0.8889        | Fm-3m                | Ba       | Nb       | Rh       |
| mp-1078969  | Ba <sub>2</sub> NbVO <sub>6</sub>  | -3.1194                    | 0.0000                 | 0.0000        | C2/m                 | Ba       | Nb       | V        |
| mp-1277332  | Ba <sub>2</sub> NbVO <sub>6</sub>  | -3.0946                    | 0.0248                 | 0.0000        | P-1                  | Ba       | Nb       | V        |
| mp-552567   | Ba <sub>2</sub> NiMoO <sub>6</sub> | -2.4493                    | 0.0000                 | 2.3514        | Fm-3m                | Ba       | Ni       | Mo       |
| mp-1147774  | Ba <sub>2</sub> NiRuO <sub>6</sub> | -2.0688                    | 0.0259                 | 0.0040        | P6 <sub>3</sub> /mmc | Ba       | Ni       | Ru       |
| mp-504723   | Ba <sub>2</sub> NiWO <sub>6</sub>  | -2.5637                    | 0.0000                 | 3.3409        | Fm-3m                | Ba       | Ni       | W        |
| mp-1521698  | BaCaFeWO <sub>6</sub>              | -2.4986                    | 0.1577                 | 0.0000        | F-43m                | Ba       | Fe       | W        |
| mp-1521804  | BaCaHfNbO <sub>6</sub>             | -3.3807                    | 0.1211                 | 0.0000        | F-43m                | Ba       | Nb       | Hf       |
| mp-1521252  | BaCaHfWO <sub>6</sub>              | -3.0561                    | 0.1809                 | 0.0000        | F-43m                | Ba       | Hf       | W        |
| mp-1516790  | BaCaHfZrO <sub>6</sub>             | -3.6351                    | 0.0950                 | 3.3163        | F-43m                | Ba       | Zr       | Hf       |
| mp-1516715  | BaCaMnWO <sub>6</sub>              | -2.6704                    | 0.0919                 | 1.8852        | F-43m                | Ba       | Mn       | W        |
| mp-1519910  | BaCaNbWO <sub>6</sub>              | -2.8180                    | 0.1584                 | 0.7631        | F-43m                | Ba       | Nb       | W        |
| mp-1519061  | BaCaZrNbO <sub>6</sub>             | -3.3139                    | 0.1137                 | 0.0000        | F-43m                | Ba       | Zr       | Nb       |
| mp-1519763  | BaCaZrWO <sub>6</sub>              | -2.9797                    | 0.1943                 | 0.0000        | F-43m                | Ba       | Zr       | W        |
| mp-1519687  | BaCrInWO <sub>6</sub>              | -2.2334                    | 0.1703                 | 0.1824        | F-43m                | Ba       | Cr       | W        |
| mp-1519016  | BaEuHfNbO <sub>6</sub>             | -3.4131                    | 0.0521                 | 0.0000        | F-43m                | Ba       | Nb       | Hf       |
| mp-1517277  | BaEuHfZrO <sub>6</sub>             | -3.6694                    | 0.0440                 | 0.4266        | F-43m                | Ba       | Zr       | Hf       |
| mp-1523280  | BaEuZrNbO <sub>6</sub>             | -3.3494                    | 0.0601                 | 0.0000        | F-43m                | Ba       | Zr       | Nb       |
| mp-1518724  | BaGdCrFeO <sub>6</sub>             | -5.1536                    | 0.0000                 | 0.0000        | F-43m                | Ba       | Cr       | Fe       |
| mp-1516647  | BaHfNbInO <sub>6</sub>             | -3.0663                    | 0.0968                 | 0.6228        | F-43m                | Ba       | Nb       | Hf       |
| mp-1522487  | BaHfTiInO <sub>6</sub>             | -3.1044                    | 0.1905                 | 0.0000        | F-43m                | Ba       | Hf       | Ti       |

| Material ID | Formula    | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|------------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-1522067  | BaInFeWO6  | -2.0893                    | 0.1948                 | 0.0000        | F-43m       | Ba       | Fe       | W        |
| mp-1228112  | BaLaCoIrO6 | -2.2562                    | 0.0773                 | 0.0000        | P1          | Ba       | Ir       | Co       |
| mp-1229299  | BaLaCrMoO6 | -2.7249                    | 0.1363                 | 0.0000        | F-43m       | Ba       | Cr       | Mo       |
| mp-40356    | BaLaMnMoO6 | -2.6442                    | 0.1936                 | 0.3824        | F-43m       | Ba       | Mn       | Mo       |
| mp-1227855  | BaLaMnRuO6 | -2.5000                    | 0.1909                 | 0.0000        | F-43m       | Ba       | Mn       | Ru       |
| mp-1272796  | BaLaMnRuO6 | -2.5673                    | 0.1236                 | 0.3210        | Cm          | Ba       | Mn       | Ru       |
| mp-690567   | BaLaMnRuO6 | -2.5412                    | 0.1497                 | 0.0000        | Pc          | Ba       | Mn       | Ru       |
| mp-1227851  | BaLaNbCrO6 | -3.0666                    | 0.1270                 | 0.0000        | F-43m       | Ba       | Cr       | Nb       |
| mp-622618   | BaLaNiRuO6 | -2.3727                    | 0.0394                 | 0.0000        | I-4         | Ba       | Ru       | Ni       |
| mp-41283    | BaLaTaMnO6 | -3.1866                    | 0.1239                 | 1.0025        | F-43m       | Ba       | Mn       | Ta       |
| mp-1227861  | BaLaTiCrO6 | -3.2908                    | 0.0536                 | 2.2522        | F-43m       | Ba       | Cr       | Ti       |
| mp-39208    | BaLaTiCrO6 | -3.2974                    | 0.0471                 | 1.2250        | Pmma        | Ba       | Cr       | Ti       |
| mp-1227842  | BaLaTiFeO6 | -3.1126                    | 0.0600                 | 1.7838        | F-43m       | Ba       | Fe       | Ti       |
| mp-1523111  | BaMnInWO6  | -2.1920                    | 0.1948                 | 0.0000        | F-43m       | Ba       | Mn       | W        |
| mp-1227835  | BaNaTiNbO6 | -3.1377                    | 0.0308                 | 1.5651        | F-43m       | Ba       | Nb       | Ti       |
| mp-1519828  | BaNb2CrO6  | -2.7138                    | 0.1806                 | 0.0000        | Fm-3m       | Ba       | Cr       | Nb       |
| mp-1520199  | BaNbInWO6  | -2.4200                    | 0.1862                 | 0.0000        | F-43m       | Ba       | Nb       | W        |
| mp-1227838  | BaNdFeCoO6 | -2.3337                    | 0.0785                 | 0.0000        | P4mm        | Ba       | Fe       | Co       |
| mp-1516816  | BaNdHfZrO6 | -3.5245                    | 0.1953                 | 0.0000        | F-43m       | Ba       | Zr       | Hf       |
| mp-1227874  | BaNdMnCoO6 | -2.4436                    | 0.0882                 | 0.0000        | Cmm2        | Ba       | Mn       | Co       |
| mp-1227378  | BaTiMnBiO6 | -2.4951                    | 0.1682                 | 0.0000        | F-43m       | Ba       | Mn       | Ti       |
| mp-1516399  | BaTiMnNbO6 | -2.8740                    | 0.1957                 | 0.0000        | F-43m       | Ba       | Mn       | Nb       |
| mp-1522797  | BaTiNbGaO6 | -2.9036                    | 0.1456                 | 0.6275        | F-43m       | Ba       | Nb       | Ti       |
| mp-1518364  | BaVInWO6   | -2.2842                    | 0.1632                 | 0.0000        | F-43m       | Ba       | V        | W        |
| mp-1517252  | BaZrNbInO6 | -3.0014                    | 0.0999                 | 0.6061        | F-43m       | Ba       | Zr       | Nb       |
| mp-1517256  | BaZrTiInO6 | -3.0284                    | 0.1986                 | 0.0000        | F-43m       | Ba       | Zr       | Ti       |
| mp-20286    | Ba2ReNiO6  | -2.4419                    | 0.0333                 | 0.0000        | Fm-3m       | Ba       | Re       | Ni       |

| Material ID | Formula                            | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group          | element1 | element3 | element4 |
|-------------|------------------------------------|----------------------------|------------------------|---------------|----------------------|----------|----------|----------|
| mp-1228308  | Ba <sub>2</sub> TaCoO <sub>6</sub> | -2.8118                    | 0.0733                 | 1.5328        | Fm-3m                | Ba       | Co       | Ta       |
| mp-31630    | Ba <sub>2</sub> TaCrO <sub>6</sub> | -3.1679                    | 0.0000                 | 2.2009        | P6 <sub>3</sub> /mmc | Ba       | Cr       | Ta       |
| mp-1205671  | Ba <sub>2</sub> TaFeO <sub>6</sub> | -3.0062                    | 0.0000                 | 1.8623        | Fm-3m                | Ba       | Fe       | Ta       |
| mp-1228319  | Ba <sub>2</sub> TaMnO <sub>6</sub> | -3.0491                    | 0.0285                 | 0.0000        | Fm-3m                | Ba       | Ta       | Mn       |
| mp-1291775  | Ba <sub>2</sub> TaMnO <sub>6</sub> | -3.0708                    | 0.0067                 | 1.1823        | Pnma                 | Ba       | Ta       | Mn       |
| mp-1307944  | Ba <sub>2</sub> TaMnO <sub>6</sub> | -3.0699                    | 0.0077                 | 1.1315        | P1                   | Ba       | Ta       | Mn       |
| mp-1309882  | Ba <sub>2</sub> TaMnO <sub>6</sub> | -3.0714                    | 0.0062                 | 1.1694        | I4/mmm               | Ba       | Ta       | Mn       |
| mp-1228536  | Ba <sub>2</sub> TiFeO <sub>6</sub> | -2.8169                    | 0.0151                 | 0.0000        | P3m1                 | Ba       | Fe       | Ti       |
| mp-1228358  | Ba <sub>2</sub> TiMnO <sub>6</sub> | -2.9717                    | 0.0194                 | 1.1588        | R-3m                 | Ba       | Mn       | Ti       |
| mp-1096782  | Ba <sub>2</sub> VFeO <sub>6</sub>  | -2.5927                    | 0.1008                 | 1.0389        | Fm-3m                | Ba       | Fe       | V        |
| mp-1096783  | Ba <sub>2</sub> VFeO <sub>6</sub>  | -2.5688                    | 0.1247                 | 0.6837        | P6 <sub>3</sub> /mmc | Ba       | Fe       | V        |
| mp-1079477  | Ba <sub>2</sub> VMoO <sub>6</sub>  | -2.6756                    | 0.0831                 | 0.4003        | Fm-3m                | Ba       | V        | Mo       |
| mp-15888    | Ba <sub>2</sub> YIrO <sub>6</sub>  | -2.7616                    | 0.0000                 | 0.0000        | Fm-3m                | Ba       | Ir       | Y        |
| mp-6428     | Ba <sub>2</sub> YRuO <sub>6</sub>  | -2.8311                    | 0.0000                 | 0.0903        | Fm-3m                | Ba       | Ru       | Y        |
| mp-1078457  | Ba <sub>2</sub> ZrTiO <sub>6</sub> | -3.5609                    | 0.0048                 | 2.2161        | Fm-3m                | Ba       | Zr       | Ti       |
| mp-1228086  | Ba <sub>2</sub> ZrWO <sub>6</sub>  | -3.0542                    | 0.0765                 | 0.0000        | Fm-3m                | Ba       | Zr       | W        |
| mp-1523377  | CaHfNbInO <sub>6</sub>             | -3.0042                    | 0.1815                 | 0.5367        | F-43m                | Ca       | Nb       | Hf       |
| mp-1048899  | CaLaCrCoO <sub>6</sub>             | -2.5083                    | 0.1357                 | 0.0000        | Pc                   | Ca       | Cr       | Co       |
| mp-1043895  | CaLaCrFeO <sub>6</sub>             | -2.7367                    | 0.0314                 | 0.0000        | Pc                   | Ca       | Cr       | Fe       |
| mp-1048863  | CaLaCrMoO <sub>6</sub>             | -2.8939                    | 0.0141                 | 1.5053        | Pc                   | Ca       | Cr       | Mo       |
| mp-1048901  | CaLaCrNiO <sub>6</sub>             | -2.5404                    | 0.0752                 | 0.6463        | Pc                   | Ca       | Cr       | Ni       |
| mp-1048921  | CaLaCrWO <sub>6</sub>              | -2.8907                    | 0.0615                 | 1.8526        | Pc                   | Ca       | Cr       | W        |
| mp-1043894  | CaLaFeCoO <sub>6</sub>             | -2.3995                    | 0.0477                 | 0.0000        | Pc                   | Ca       | Fe       | Co       |
| mp-1044000  | CaLaFeMoO <sub>6</sub>             | -2.7122                    | 0.0240                 | 1.2186        | Pc                   | Ca       | Fe       | Mo       |
| mp-1043874  | CaLaFeNiO <sub>6</sub>             | -2.3526                    | 0.0000                 | 0.0000        | Pc                   | Ca       | Fe       | Ni       |
| mp-1638745  | CaLaFeNiO <sub>6</sub>             | -2.3485                    | 0.0040                 | 0.3580        | P1                   | Ca       | Fe       | Ni       |
| mp-1044027  | CaLaFeWO <sub>6</sub>              | -2.7854                    | 0.0372                 | 0.2204        | P1                   | Ca       | Fe       | W        |

| Material ID | Formula    | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|------------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-1638691  | CaLaFeWO6  | -2.7882                    | 0.0345                 | 2.4710        | P1          | Ca       | Fe       | W        |
| mp-1049271  | CaLaMnCrO6 | -2.8558                    | 0.1069                 | 0.0000        | Pc          | Ca       | Cr       | Mn       |
| mp-40066    | CaLaMnFeO6 | -2.6977                    | 0.0575                 | 1.0757        | Pc          | Ca       | Fe       | Mn       |
| mp-41963    | CaLaMnFeO6 | -2.6906                    | 0.0646                 | 1.4781        | Pc          | Ca       | Fe       | Mn       |
| mp-684705   | CaLaMnMoO6 | -2.8262                    | 0.0631                 | 1.3872        | Pc          | Ca       | Mn       | Mo       |
| mp-1227173  | CaLaMnNbO6 | -3.2073                    | 0.0076                 | 2.4089        | Pc          | Ca       | Mn       | Nb       |
| mp-694885   | CaLaMnNbO6 | -3.1626                    | 0.0524                 | 2.4439        | Pc          | Ca       | Mn       | Nb       |
| mp-1227224  | CaLaNbCoO6 | -3.0533                    | 0.0095                 | 0.7135        | P1          | Ca       | Nb       | Co       |
| mp-1040806  | CaLaTaCrO6 | -3.2918                    | 0.0351                 | 2.1046        | Pc          | Ca       | Cr       | Ta       |
| mp-1043970  | CaLaTaFeO6 | -3.2172                    | 0.0000                 | 1.8972        | P1          | Ca       | Fe       | Ta       |
| mp-39159    | CaLaTiCrO6 | -3.3767                    | 0.0064                 | 2.6877        | Pc          | Ca       | Cr       | Ti       |
| mp-1043825  | CaLaTiFeO6 | -3.2005                    | 0.0040                 | 0.0000        | Pc          | Ca       | Fe       | Ti       |
| mp-40866    | CaLaTiMnO6 | -3.2612                    | 0.0886                 | 0.2488        | Pc          | Ca       | Mn       | Ti       |
| mp-1048855  | CaLaVCrO6  | -3.0607                    | 0.0129                 | 1.5655        | Pc          | Ca       | Cr       | V        |
| mp-1043924  | CaLaVFeO6  | -2.8882                    | 0.0128                 | 0.0000        | Pc          | Ca       | Fe       | V        |
| mp-1227087  | CaLaVFeO6  | -2.7651                    | 0.1358                 | 1.0776        | F-43m       | Ca       | Fe       | V        |
| mp-684837   | CaLaVFeO6  | -2.8701                    | 0.0309                 | 0.1109        | P1          | Ca       | Fe       | V        |
| mp-1227209  | CaMnFeBiO6 | -2.0966                    | 0.0142                 | 0.7049        | Pc          | Ca       | Fe       | Mn       |
| mp-1342876  | CaMnIrO6   | -1.7609                    | 0.0842                 | 0.0000        | Pc          | Ca       | Mn       | Ir       |
| mp-1290713  | CaMnWO6    | -2.3135                    | 0.1114                 | 0.8557        | Pc          | Ca       | Mn       | W        |
| mp-1519219  | CaNdHfFeO6 | -3.1186                    | 0.1801                 | 2.0833        | F-43m       | Ca       | Fe       | Hf       |
| mp-1521542  | CaNdHfTiO6 | -3.5245                    | 0.1904                 | 0.0000        | F-43m       | Ca       | Hf       | Ti       |
| mp-41742    | CaNdTiMnO6 | -3.0850                    | 0.1748                 | 0.0000        | Pc          | Ca       | Mn       | Ti       |
| mp-1522086  | CaNdZrCrO6 | -3.2070                    | 0.1991                 | 2.4074        | F-43m       | Ca       | Cr       | Zr       |
| mp-1200872  | CaTi2FeO6  | -3.1236                    | 0.0193                 | 1.3812        | P4_2/nmc    | Ca       | Fe       | Ti       |
| mp-1194928  | CaTi2MnO6  | -3.2375                    | 0.0179                 | 1.5800        | P4_2mc      | Ca       | Mn       | Ti       |
| mp-1520239  | CaTiNbInO6 | -2.8622                    | 0.1796                 | 0.0000        | F-43m       | Ca       | Nb       | Ti       |

| Material ID | Formula    | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|------------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-1517467  | CaZrNbInO6 | -2.9280                    | 0.1959                 | 0.5202        | F-43m       | Ca       | Zr       | Nb       |
| mp-1287594  | Ca2CoIrO6  | -2.1264                    | 0.0000                 | 0.0000        | P1          | Ca       | Co       | Ir       |
| mp-1310548  | Ca2CoIrO6  | -2.0968                    | 0.0296                 | 0.0000        | P1          | Ca       | Co       | Ir       |
| mp-1378169  | Ca2CoRhO6  | -1.9180                    | 0.1493                 | 0.0746        | C2          | Ca       | Co       | Rh       |
| mp-1045281  | Ca2CoWO6   | -2.5921                    | 0.0249                 | 0.0000        | P1          | Ca       | Co       | W        |
| mp-1398590  | Ca2CoWO6   | -2.4928                    | 0.1241                 | 1.4930        | Cm          | Ca       | Co       | W        |
| mp-19161    | Ca2CoWO6   | -2.6169                    | 0.0000                 | 1.2029        | P2_1/c      | Ca       | Co       | W        |
| mp-1367307  | Ca2CrIrO6  | -2.4017                    | 0.0000                 | 0.0000        | P2_1/c      | Ca       | Ir       | Cr       |
| mp-1045238  | Ca2CrWO6   | -2.7686                    | 0.0342                 | 1.6952        | P1          | Ca       | W        | Cr       |
| mp-1272359  | Ca2CrWO6   | -2.7694                    | 0.0333                 | 1.8837        | P-1         | Ca       | W        | Cr       |
| mp-1282800  | Ca2CrWO6   | -2.7877                    | 0.0151                 | 1.7595        | P2_1/c      | Ca       | W        | Cr       |
| mp-1445093  | Ca2CrWO6   | -2.7683                    | 0.0345                 | 1.6023        | P1          | Ca       | W        | Cr       |
| mp-1306506  | Ca2FeIrO6  | -2.2419                    | 0.0000                 | 0.0000        | P2_1/c      | Ca       | Fe       | Ir       |
| mp-18783    | Ca2FeMoO6  | -2.5481                    | 0.0040                 | 1.1193        | P2_1/c      | Ca       | Fe       | Mo       |
| mp-31763    | Ca2FeReO6  | -2.5547                    | 0.0000                 | 0.0000        | P2_1/c      | Ca       | Re       | Fe       |
| mp-1048318  | Ca2FeWO6   | -2.6502                    | 0.0223                 | 3.2631        | P1          | Ca       | Fe       | W        |
| mp-1275487  | Ca2FeWO6   | -2.6725                    | 0.0000                 | 3.0012        | P2_1/c      | Ca       | Fe       | W        |
| mp-19059    | Ca2FeWO6   | -2.4865                    | 0.1860                 | 0.0000        | Fm-3m       | Ca       | Fe       | W        |
| mp-19864    | Ca2FeWO6   | -2.5468                    | 0.1257                 | 2.7594        | Imm2        | Ca       | Fe       | W        |
| mp-619611   | Ca2FeWO6   | -2.4855                    | 0.1870                 | 2.5738        | R-3m        | Ca       | Fe       | W        |
| mp-623098   | Ca2FeWO6   | -2.5011                    | 0.1714                 | 1.8750        | Pmm2        | Ca       | Fe       | W        |
| mp-1523252  | Ca2HfNbO6  | -3.3322                    | 0.1863                 | 0.0000        | Fm-3m       | Ca       | Nb       | Hf       |
| mp-1516534  | Ca2HfTiO6  | -3.5724                    | 0.1161                 | 2.4866        | Fm-3m       | Ca       | Hf       | Ti       |
| mp-1522139  | Ca2HfZrO6  | -3.5721                    | 0.1734                 | 3.4088        | Fm-3m       | Ca       | Zr       | Hf       |
| mp-1227404  | Ca2IrRhO6  | -2.1062                    | 0.0000                 | 0.0000        | C2/m        | Ca       | Rh       | Ir       |
| mp-1369951  | Ca2IrWO6   | -2.3918                    | 0.1557                 | 1.1277        | P2_1/c      | Ca       | Ir       | W        |
| mp-39440    | Ca2MnIrO6  | -2.2980                    | 0.0207                 | 0.0000        | P2_1/c      | Ca       | Mn       | Ir       |

| Material ID | Formula                            | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group        | element1 | element3 | element4 |
|-------------|------------------------------------|----------------------------|------------------------|---------------|--------------------|----------|----------|----------|
| mp-1227519  | Ca <sub>2</sub> MnNbO <sub>6</sub> | -3.0011                    | 0.0131                 | 0.0000        | P2 <sub>1</sub> /c | Ca       | Nb       | Mn       |
| mp-1326305  | Ca <sub>2</sub> MnNiO <sub>6</sub> | -1.9918                    | 0.1887                 | 0.0000        | C2                 | Ca       | Mn       | Ni       |
| mp-1106155  | Ca <sub>2</sub> MnReO <sub>6</sub> | -2.6560                    | 0.0008                 | 0.1472        | P2 <sub>1</sub> /c | Ca       | Re       | Mn       |
| mp-1227575  | Ca <sub>2</sub> MnRuO <sub>6</sub> | -2.3942                    | 0.0000                 | 0.0000        | P2 <sub>1</sub> /c | Ca       | Mn       | Ru       |
| mp-1045241  | Ca <sub>2</sub> MnWO <sub>6</sub>  | -2.7570                    | 0.0206                 | 2.6496        | P1                 | Ca       | Mn       | W        |
| mp-1391233  | Ca <sub>2</sub> MnWO <sub>6</sub>  | -2.7554                    | 0.0222                 | 2.5151        | P1                 | Ca       | Mn       | W        |
| mp-19108    | Ca <sub>2</sub> MnWO <sub>6</sub>  | -2.7775                    | 0.0000                 | 2.1817        | P2 <sub>1</sub> /c | Ca       | Mn       | W        |
| mp-1047241  | Ca <sub>2</sub> MoWO <sub>6</sub>  | -2.6374                    | 0.0247                 | 1.9254        | P-1                | Ca       | W        | Mo       |
| mp-1347979  | Ca <sub>2</sub> MoWO <sub>6</sub>  | -2.6615                    | 0.0006                 | 1.5506        | P2 <sub>1</sub> /c | Ca       | W        | Mo       |
| mp-1214115  | Ca <sub>2</sub> NbCrO <sub>6</sub> | -3.1158                    | 0.0000                 | 2.7637        | P2 <sub>1</sub> /c | Ca       | Nb       | Cr       |
| mp-1214121  | Ca <sub>2</sub> NbFeO <sub>6</sub> | -2.9417                    | 0.0000                 | 2.0871        | P2 <sub>1</sub> /c | Ca       | Nb       | Fe       |
| mp-1047142  | Ca <sub>2</sub> NiIrO <sub>6</sub> | -2.0944                    | 0.0000                 | 0.0000        | P2 <sub>1</sub> /c | Ca       | Ni       | Ir       |
| mp-1397424  | Ca <sub>2</sub> NiWO <sub>6</sub>  | -2.5017                    | 0.1065                 | 0.5975        | C2                 | Ca       | Ni       | W        |
| mp-19114    | Ca <sub>2</sub> NiWO <sub>6</sub>  | -2.6082                    | 0.0000                 | 2.9330        | P2 <sub>1</sub> /c | Ca       | Ni       | W        |
| mp-556384   | Ca <sub>2</sub> NiWO <sub>6</sub>  | -2.5065                    | 0.1017                 | 1.5013        | P2 <sub>1</sub> /c | Ca       | Ni       | W        |
| mp-1377373  | Ca <sub>2</sub> TaCoO <sub>6</sub> | -2.8891                    | 0.0373                 | 0.3612        | P2 <sub>1</sub> /m | Ca       | Co       | Ta       |
| mp-1214103  | Ca <sub>2</sub> TaCrO <sub>6</sub> | -3.2333                    | 0.0000                 | 2.7922        | P2 <sub>1</sub> /c | Ca       | Ta       | Cr       |
| mp-1345721  | Ca <sub>2</sub> TaCrO <sub>6</sub> | -3.2102                    | 0.0232                 | 2.6352        | P2 <sub>1</sub> /m | Ca       | Ta       | Cr       |
| mp-1341944  | Ca <sub>2</sub> TaFeO <sub>6</sub> | -3.0356                    | 0.0050                 | 1.2064        | P2 <sub>1</sub> /m | Ca       | Fe       | Ta       |
| mp-558810   | Ca <sub>2</sub> TaMnO <sub>6</sub> | -3.1131                    | 0.0086                 | 0.0000        | P2 <sub>1</sub> /m | Ca       | Ta       | Mn       |
| mp-1366540  | Ca <sub>2</sub> TaMoO <sub>6</sub> | -3.0838                    | 0.0000                 | 1.4283        | P2 <sub>1</sub> /m | Ca       | Ta       | Mo       |
| mp-1367927  | Ca <sub>2</sub> TaNiO <sub>6</sub> | -2.7718                    | 0.0863                 | 0.0000        | P2 <sub>1</sub> /m | Ca       | Ta       | Ni       |
| mp-1373655  | Ca <sub>2</sub> TaTiO <sub>6</sub> | -3.4458                    | 0.0045                 | 0.0000        | P2 <sub>1</sub> /m | Ca       | Ta       | Ti       |
| mp-1373426  | Ca <sub>2</sub> TaVO <sub>6</sub>  | -3.2546                    | 0.0000                 | 0.4315        | P2 <sub>1</sub> /m | Ca       | Ta       | V        |
| mp-1367958  | Ca <sub>2</sub> TaWO <sub>6</sub>  | -3.0342                    | 0.0726                 | 2.1053        | P2 <sub>1</sub> /c | Ca       | Ta       | W        |
| mp-1372505  | Ca <sub>2</sub> TaWO <sub>6</sub>  | -3.0086                    | 0.0982                 | 0.0000        | P2 <sub>1</sub> /m | Ca       | Ta       | W        |
| mp-1443834  | Ca <sub>2</sub> TaWO <sub>6</sub>  | -3.0057                    | 0.1011                 | 2.2589        | R3                 | Ca       | Ta       | W        |

| Material ID | Formula                            | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group        | element1 | element3 | element4 |
|-------------|------------------------------------|----------------------------|------------------------|---------------|--------------------|----------|----------|----------|
| mp-1227758  | Ca <sub>2</sub> TiCrO <sub>6</sub> | -3.0899                    | 0.0299                 | 1.1492        | P2 <sub>1</sub> /c | Ca       | Ti       | Cr       |
| mp-1340170  | Ca <sub>2</sub> TiIrO <sub>6</sub> | -2.8119                    | 0.0171                 | 0.0000        | P2 <sub>1</sub> /c | Ca       | Ti       | Ir       |
| mp-1080507  | Ca <sub>2</sub> TiMnO <sub>6</sub> | -3.0255                    | 0.0203                 | 0.8750        | I4/m               | Ca       | Mn       | Ti       |
| mp-1227706  | Ca <sub>2</sub> TiMnO <sub>6</sub> | -3.0405                    | 0.0053                 | 1.9470        | P2 <sub>1</sub> /c | Ca       | Mn       | Ti       |
| mp-1227718  | Ca <sub>2</sub> TiRuO <sub>6</sub> | -2.8885                    | 0.0029                 | 0.0000        | P2 <sub>1</sub> /c | Ca       | Ti       | Ru       |
| mp-1367801  | Ca <sub>2</sub> TiWO <sub>6</sub>  | -3.0820                    | 0.0319                 | 0.3989        | P2 <sub>1</sub> /c | Ca       | Ti       | W        |
| mp-1384639  | Ca <sub>2</sub> TiWO <sub>6</sub>  | -2.9814                    | 0.1325                 | 0.0000        | Cm                 | Ca       | Ti       | W        |
| mp-1379764  | Ca <sub>2</sub> VIrO <sub>6</sub>  | -2.4848                    | 0.0923                 | 0.4312        | P2 <sub>1</sub> /c | Ca       | Ir       | V        |
| mp-1227834  | Ca <sub>2</sub> VMoO <sub>6</sub>  | -2.7242                    | 0.0543                 | 0.6962        | P2 <sub>1</sub> /c | Ca       | V        | Mo       |
| mp-1047139  | Ca <sub>2</sub> VWO <sub>6</sub>   | -2.7895                    | 0.0509                 | 0.8829        | P-1                | Ca       | V        | W        |
| mp-1274301  | Ca <sub>2</sub> VWO <sub>6</sub>   | -2.7926                    | 0.0479                 | 1.0210        | P-1                | Ca       | V        | W        |
| mp-1296431  | Ca <sub>2</sub> VWO <sub>6</sub>   | -2.8138                    | 0.0267                 | 0.0000        | P2 <sub>1</sub> /c | Ca       | V        | W        |
| mp-1521637  | Ca <sub>2</sub> ZrFeO <sub>6</sub> | -2.7999                    | 0.1495                 | 0.0000        | Fm-3m              | Ca       | Fe       | Zr       |
| mp-1517008  | Ca <sub>2</sub> ZrNbO <sub>6</sub> | -3.2531                    | 0.1900                 | 0.0000        | Fm-3m              | Ca       | Nb       | Zr       |
| mp-1518628  | Ca <sub>2</sub> ZrTiO <sub>6</sub> | -3.4794                    | 0.1337                 | 2.3933        | Fm-3m              | Ca       | Zr       | Ti       |
| mp-1652492  | Ca <sub>3</sub> CoRhO <sub>6</sub> | -2.3654                    | 0.0194                 | 0.0000        | P1                 | Ca       | Co       | Rh       |
| mp-1666380  | Ca <sub>3</sub> CoRhO <sub>6</sub> | -2.3666                    | 0.0182                 | 0.0245        | P-1                | Ca       | Co       | Rh       |
| mp-1983384  | Ca <sub>3</sub> CoRhO <sub>6</sub> | -2.3657                    | 0.0192                 | 0.0000        | P1                 | Ca       | Co       | Rh       |
| mp-1986469  | Ca <sub>3</sub> CoRhO <sub>6</sub> | -2.3658                    | 0.0190                 | 0.0000        | P1                 | Ca       | Co       | Rh       |
| mp-551544   | Ca <sub>3</sub> CoRhO <sub>6</sub> | -2.3664                    | 0.0184                 | 0.2555        | P1                 | Ca       | Co       | Rh       |
| mp-551706   | Ca <sub>3</sub> CoRhO <sub>6</sub> | -2.3280                    | 0.0568                 | 0.7604        | R-3c               | Ca       | Co       | Rh       |
| mp-18843    | Ca <sub>3</sub> FeRhO <sub>6</sub> | -2.4607                    | 0.0162                 | 0.8859        | R-3c               | Ca       | Fe       | Rh       |
| mp-1232815  | Ca <sub>3</sub> MnCoO <sub>6</sub> | -2.6075                    | 0.0000                 | 2.1034        | P1                 | Ca       | Co       | Mn       |
| mp-561610   | Ca <sub>3</sub> MnCoO <sub>6</sub> | -2.6075                    | 0.0000                 | 0.4421        | P1                 | Ca       | Co       | Mn       |
| mp-22533    | Ca <sub>3</sub> MnNiO <sub>6</sub> | -2.5753                    | 0.0009                 | 0.0000        | R-3c               | Ca       | Mn       | Ni       |
| mp-997184   | Ca <sub>3</sub> TiNiO <sub>6</sub> | -2.9818                    | 0.0582                 | 3.1552        | R-3c               | Ca       | Ni       | Ti       |
| mp-1372511  | CaCoWO <sub>6</sub>                | -1.9933                    | 0.1815                 | 0.0000        | Pc                 | Ca       | Co       | W        |



| Material ID | Formula    | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|------------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-1522808  | CaEuHfNbO6 | -3.3592                    | 0.1330                 | 0.0000        | F-43m       | Ca       | Nb       | Hf       |
| mp-1519554  | CaEuHfTiO6 | -3.5972                    | 0.0925                 | 0.0000        | F-43m       | Ca       | Hf       | Ti       |
| mp-1519493  | CaEuHfVO6  | -3.2651                    | 0.1403                 | 0.0000        | F-43m       | Ca       | Hf       | V        |
| mp-1522072  | CaEuHfZrO6 | -3.6019                    | 0.1286                 | 0.3977        | F-43m       | Ca       | Zr       | Hf       |
| mp-1517301  | CaEuMnWO6  | -2.6462                    | 0.1149                 | 0.2840        | F-43m       | Ca       | Mn       | W        |
| mp-1521619  | CaEuNbFeO6 | -2.8605                    | 0.1108                 | 0.0000        | F-43m       | Ca       | Fe       | Nb       |
| mp-1521618  | CaEuZrMnO6 | -3.0254                    | 0.1822                 | 0.0000        | F-43m       | Ca       | Zr       | Mn       |
| mp-1519002  | CaEuZrNbO6 | -3.2841                    | 0.1439                 | 0.0000        | F-43m       | Ca       | Zr       | Nb       |
| mp-1518717  | CaEuZrTiO6 | -3.5078                    | 0.1066                 | 0.0000        | F-43m       | Ca       | Zr       | Ti       |
| mp-1518352  | CaEuZrVO6  | -3.1782                    | 0.1531                 | 0.0000        | F-43m       | Ca       | Zr       | V        |
| mp-1343817  | CaFeReO6   | -2.2566                    | 0.0434                 | 0.6851        | Pc          | Ca       | Re       | Fe       |
| mp-1385922  | CaFeWO6    | -2.1410                    | 0.1506                 | 0.0000        | F-43m       | Ca       | Fe       | W        |
| mp-1391360  | CaFeWO6    | -2.1416                    | 0.1500                 | 0.0000        | Cm          | Ca       | Fe       | W        |
| mp-1237349  | Dy2MnCoO6  | -2.6561                    | 0.1969                 | 0.3918        | P2_1/c      | Dy       | Co       | Mn       |
| mp-1213197  | Dy2NiRuO6  | -2.6507                    | 0.0327                 | 0.0000        | P2_1/c      | Dy       | Ni       | Ru       |
| mp-1225428  | Dy2VFeO6   | -3.0615                    | 0.0230                 | 1.3296        | P2_1/c      | Dy       | Fe       | V        |
| mp-1213340  | Eu2CoRuO6  | -2.3392                    | 0.0750                 | 0.0000        | P2_1/c      | Eu       | Co       | Ru       |
| mp-1520430  | Eu2CoWO6   | -2.4780                    | 0.1255                 | 0.0000        | Fm-3m       | Eu       | Co       | W        |
| mp-1521041  | Eu2FeWO6   | -2.5211                    | 0.1138                 | 0.0000        | Fm-3m       | Eu       | Fe       | W        |
| mp-1518318  | Eu2HfFeO6  | -3.0686                    | 0.1562                 | 0.0000        | Fm-3m       | Eu       | Hf       | Fe       |
| mp-1518357  | Eu2HfNbO6  | -3.3894                    | 0.0423                 | 0.0000        | Fm-3m       | Eu       | Hf       | Nb       |
| mp-1521653  | Eu2HfTiO6  | -3.6221                    | 0.0170                 | 0.0035        | Fm-3m       | Eu       | Hf       | Ti       |
| mp-1518592  | Eu2HfVO6   | -3.3054                    | 0.0567                 | 0.0000        | Fm-3m       | Eu       | Hf       | V        |
| mp-1518017  | Eu2HfWO6   | -3.0743                    | 0.1597                 | 0.0000        | Fm-3m       | Eu       | Hf       | W        |
| mp-1520691  | Eu2HfZrO6  | -3.6334                    | 0.0466                 | 0.3481        | Fm-3m       | Eu       | Hf       | Zr       |
| mp-1225753  | Eu2MnRhO6  | -2.4150                    | 0.0907                 | 0.0000        | P2_1/c      | Eu       | Mn       | Rh       |
| mp-1516440  | Eu2MnWO6   | -2.6749                    | 0.0698                 | 0.3876        | Fm-3m       | Eu       | Mn       | W        |

| Material ID | Formula    | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|------------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-1516596  | Eu2NbFeO6  | -2.8867                    | 0.0830                 | 0.0000        | Fm-3m       | Eu       | Fe       | Nb       |
| mp-1519174  | Eu2NbWO6   | -2.8139                    | 0.1627                 | 0.0000        | Fm-3m       | Eu       | Nb       | W        |
| mp-1516887  | Eu2NiWO6   | -2.5163                    | 0.0826                 | 0.4430        | Fm-3m       | Eu       | Ni       | W        |
| mp-1521061  | Eu2TiNbO6  | -3.3433                    | 0.0077                 | 0.0000        | Fm-3m       | Eu       | Nb       | Ti       |
| mp-1520585  | Eu2VWO6    | -2.7339                    | 0.0806                 | 0.3942        | Fm-3m       | Eu       | V        | W        |
| mp-1520930  | Eu2ZrCrO6  | -3.1661                    | 0.1069                 | 0.0000        | Fm-3m       | Eu       | Cr       | Zr       |
| mp-1519374  | Eu2ZrFeO6  | -2.9746                    | 0.1883                 | 0.0000        | Fm-3m       | Eu       | Fe       | Zr       |
| mp-1516495  | Eu2ZrNbO6  | -3.3173                    | 0.0746                 | 0.0000        | Fm-3m       | Eu       | Nb       | Zr       |
| mp-1520462  | Eu2ZrTiO6  | -3.5371                    | 0.0623                 | 0.0034        | Fm-3m       | Eu       | Zr       | Ti       |
| mp-1523032  | Eu2ZrVO6   | -3.2176                    | 0.0974                 | 0.0000        | Fm-3m       | Eu       | Zr       | V        |
| mp-1520588  | Eu2ZrWO6   | -2.9843                    | 0.1879                 | 0.0000        | Fm-3m       | Eu       | Zr       | W        |
| mp-1519362  | EuHfCrBiO6 | -2.7592                    | 0.1746                 | 0.0000        | F-43m       | Eu       | Cr       | Hf       |
| mp-1224833  | Gd2MnCoO6  | -2.8105                    | 0.0000                 | 0.9184        | P2_1/c      | Gd       | Co       | Mn       |
| mp-1212655  | Gd2MnNiO6  | -2.8014                    | 0.0000                 | 0.0000        | P2_1/c      | Gd       | Mn       | Ni       |
| mp-1212785  | Gd2NiRuO6  | -2.6016                    | 0.0271                 | 0.0000        | P2_1/c      | Gd       | Ni       | Ru       |
| mp-1225231  | Gd2VFeO6   | -3.0350                    | 0.0013                 | 1.6075        | P2_1/c      | Gd       | Fe       | V        |
| mp-1519547  | KCaNbWO6   | -2.6291                    | 0.1333                 | 0.0000        | F-43m       | K        | Nb       | W        |
| mp-1520344  | KCaZrNbO6  | -3.1861                    | 0.0751                 | 2.2430        | F-43m       | K        | Zr       | Nb       |
| mp-1521506  | KEuHfFeO6  | -2.5958                    | 0.1860                 | 0.0000        | F-43m       | K        | Fe       | Hf       |
| mp-1521954  | KEuHfTiO6  | -3.2929                    | 0.1503                 | 0.0000        | F-43m       | K        | Hf       | Ti       |
| mp-1518234  | KEuHfZrO6  | -3.3276                    | 0.1664                 | 0.0000        | F-43m       | K        | Zr       | Hf       |
| mp-1523222  | KFeBiWO6   | -1.7819                    | 0.1846                 | 0.0000        | F-43m       | K        | Fe       | W        |
| mp-1521336  | KGdHfTiO6  | -3.3935                    | 0.1822                 | 0.8886        | F-43m       | K        | Hf       | Ti       |
| mp-1522209  | KHfNbBiO6  | -2.7263                    | 0.1807                 | 0.0000        | F-43m       | K        | Nb       | Hf       |
| mp-1516503  | KHfNbInO6  | -2.7755                    | 0.1889                 | 0.0000        | F-43m       | K        | Nb       | Hf       |
| mp-1521210  | KHfZrBiO6  | -2.8717                    | 0.1730                 | 0.8020        | F-43m       | K        | Zr       | Hf       |
| mp-1516194  | KLaCoWO6   | -2.3501                    | 0.1804                 | 0.0000        | F-43m       | K        | Co       | W        |

| Material ID | Formula   | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|-----------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-1523210  | KLaFeWO6  | -2.4268                    | 0.1457                 | 0.0000        | F-43m       | K        | Fe       | W        |
| mp-1522697  | KLaHfCrO6 | -3.0130                    | 0.1747                 | 0.0000        | F-43m       | K        | Cr       | Hf       |
| mp-1518711  | KLaHfFeO6 | -2.8129                    | 0.1169                 | 0.0000        | F-43m       | K        | Fe       | Hf       |
| mp-1520833  | KLaHfNbO6 | -3.3111                    | 0.1049                 | 0.0000        | F-43m       | K        | Nb       | Hf       |
| mp-1516609  | KLaHfTiO6 | -3.5040                    | 0.0952                 | 2.3179        | F-43m       | K        | Hf       | Ti       |
| mp-1105566  | KLaMnWO6  | -2.6865                    | 0.0000                 | 2.4917        | P2_1        | K        | Mn       | W        |
| mp-1520582  | KLaMnWO6  | -2.5746                    | 0.1119                 | 1.1266        | F-43m       | K        | Mn       | W        |
| mp-1522297  | KLaNbCrO6 | -2.9387                    | 0.0856                 | 2.0549        | F-43m       | K        | Cr       | Nb       |
| mp-1522180  | KLaNbFeO6 | -2.7651                    | 0.0874                 | 1.7118        | F-43m       | K        | Fe       | Nb       |
| mp-1520198  | KLaNbNiO6 | -2.5255                    | 0.1465                 | 0.0000        | F-43m       | K        | Nb       | Ni       |
| mp-1516686  | KLaNbWO6  | -2.7293                    | 0.1953                 | 0.4343        | F-43m       | K        | Nb       | W        |
| mp-1517874  | KLaNiWO6  | -2.4090                    | 0.1107                 | 2.8449        | F-43m       | K        | Ni       | W        |
| mp-1521711  | KLaTiNbO6 | -3.2391                    | 0.0457                 | 0.0000        | F-43m       | K        | Nb       | Ti       |
| mp-1521904  | KLaVWO6   | -2.6434                    | 0.1062                 | 0.5106        | F-43m       | K        | V        | W        |
| mp-1517088  | KLaZrCrO6 | -2.9300                    | 0.1897                 | 0.0000        | F-43m       | K        | Cr       | Zr       |
| mp-1519849  | KLaZrFeO6 | -2.7312                    | 0.1287                 | 0.0000        | F-43m       | K        | Fe       | Zr       |
| mp-1522104  | KLaZrNbO6 | -3.2380                    | 0.1079                 | 0.0000        | F-43m       | K        | Zr       | Nb       |
| mp-1520792  | KLaZrTiO6 | -3.4186                    | 0.1075                 | 2.2347        | F-43m       | K        | Zr       | Ti       |
| mp-1520546  | KLaZrVO6  | -3.0741                    | 0.1634                 | 0.0000        | F-43m       | K        | Zr       | V        |
| mp-1522301  | KMnBiWO6  | -1.8992                    | 0.1775                 | 0.0000        | F-43m       | K        | Mn       | W        |
| mp-1516600  | KNdHfTiO6 | -3.4365                    | 0.1289                 | 2.2033        | F-43m       | K        | Hf       | Ti       |
| mp-1521021  | KNdHfZrO6 | -3.4517                    | 0.1575                 | 2.6382        | F-43m       | K        | Zr       | Hf       |
| mp-1105240  | KNdMnWO6  | -2.6512                    | 0.0020                 | 2.6477        | P2_1        | K        | Mn       | W        |
| mp-1520724  | KNdZrFeO6 | -2.6637                    | 0.1586                 | 0.0000        | F-43m       | K        | Fe       | Zr       |
| mp-1517399  | KNdZrNbO6 | -3.1543                    | 0.1560                 | 0.0000        | F-43m       | K        | Zr       | Nb       |
| mp-1523257  | KNdZrTiO6 | -3.3488                    | 0.1405                 | 2.1104        | F-43m       | K        | Zr       | Ti       |
| mp-1519358  | KNiBiWO6  | -1.7303                    | 0.1835                 | 1.3429        | F-43m       | K        | Ni       | W        |

| Material ID | Formula   | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|-----------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-1520212  | KVBiWO6   | -1.9858                    | 0.1810                 | 0.0000        | F-43m       | K        | V        | W        |
| mp-1521496  | KZrMnBiO6 | -2.2560                    | 0.1577                 | 1.1158        | F-43m       | K        | Mn       | Zr       |
| mp-1518641  | KZrNbBiO6 | -2.6521                    | 0.1916                 | 0.0000        | F-43m       | K        | Zr       | Nb       |
| mp-18862    | La2CoIrO6 | -2.5307                    | 0.0494                 | 0.6160        | P2_1/c      | La       | Co       | Ir       |
| mp-1223259  | La2CoNiO6 | -2.5728                    | 0.0000                 | 0.9972        | R-3         | La       | Co       | Ni       |
| mp-1275360  | La2CoNiO6 | -2.5321                    | 0.0407                 | 0.0390        | P-1         | La       | Co       | Ni       |
| mp-1211319  | La2CoRuO6 | -2.6476                    | 0.0000                 | 0.0000        | P2_1/c      | La       | Co       | Ru       |
| mp-1223369  | La2CrFeO6 | -3.0199                    | 0.0049                 | 0.3097        | P2_1/c      | La       | Cr       | Fe       |
| mp-1223257  | La2CrMoO6 | -3.0065                    | 0.0317                 | 0.0000        | I4/m        | La       | Cr       | Mo       |
| mp-1223373  | La2FeCoO6 | -2.6998                    | 0.0266                 | 0.0000        | P2_1/c      | La       | Co       | Fe       |
| mp-1223335  | La2FeNiO6 | -2.6623                    | 0.0003                 | 0.0000        | P2_1/c      | La       | Fe       | Ni       |
| mp-1078675  | La2HfNiO6 | -3.2478                    | 0.1880                 | 2.0434        | Fm-3m       | La       | Hf       | Ni       |
| mp-19208    | La2MnCoO6 | -3.2242                    | 0.0000                 | 0.0000        | P2_1/c      | La       | Co       | Mn       |
| mp-1223342  | La2MnCrO6 | -3.0808                    | 0.0851                 | 0.7624        | P2_1/c      | La       | Cr       | Mn       |
| mp-1666963  | La2MnCrO6 | -3.0835                    | 0.0824                 | 0.9733        | P-1         | La       | Cr       | Mn       |
| mp-1079517  | La2MnNiO6 | -2.7740                    | 0.0973                 | 1.2797        | Fm-3m       | La       | Ni       | Mn       |
| mp-1188812  | La2MnNiO6 | -2.8285                    | 0.0427                 | 0.0000        | P2_1/c      | La       | Ni       | Mn       |
| mp-1223253  | La2MnNiO6 | -2.8236                    | 0.0477                 | 1.5805        | R-3         | La       | Ni       | Mn       |
| mp-1223338  | La2MnRhO6 | -2.7626                    | 0.1139                 | 0.3777        | P2_1/c      | La       | Rh       | Mn       |
| mp-1223380  | La2MnRuO6 | -2.7771                    | 0.1260                 | 0.0000        | P2_1/c      | La       | Mn       | Ru       |
| mp-560369   | La2MnVO6  | -3.1588                    | 0.0484                 | 0.0000        | P2_1/c      | La       | Mn       | V        |
| mp-1223155  | La2NbCrO6 | -3.1323                    | 0.1890                 | 0.0000        | I4/m        | La       | Cr       | Nb       |
| mp-1211522  | La2NbNiO6 | -3.0652                    | 0.1721                 | 0.2076        | P2_1/c      | La       | Nb       | Ni       |
| mp-20964    | La2NiIrO6 | -2.5641                    | 0.0052                 | 0.0000        | P2_1        | La       | Ni       | Ir       |
| mp-1223214  | La2NiRhO6 | -2.5722                    | 0.0000                 | 0.0000        | P2_1/c      | La       | Ni       | Rh       |
| mp-19176    | La2NiRuO6 | -2.6334                    | 0.0000                 | 0.0000        | P2_1/c      | La       | Ni       | Ru       |
| mp-1087477  | La2TcNiO6 | -2.6544                    | 0.0922                 | 0.0000        | Fm-3m       | La       | Ni       | Tc       |

| Material ID | Formula                            | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group        | element1 | element3 | element4 |
|-------------|------------------------------------|----------------------------|------------------------|---------------|--------------------|----------|----------|----------|
| mp-1078893  | La <sub>2</sub> TiCoO <sub>6</sub> | -3.1401                    | 0.1978                 | 0.0000        | Fm-3m              | La       | Co       | Ti       |
| mp-1211314  | La <sub>2</sub> TiCoO <sub>6</sub> | -3.3241                    | 0.0138                 | 0.5484        | P-1                | La       | Co       | Ti       |
| mp-1225793  | La <sub>2</sub> TiCrO <sub>6</sub> | -3.3934                    | 0.0874                 | 0.0000        | R-3                | La       | Cr       | Ti       |
| mp-1276293  | La <sub>2</sub> TiCrO <sub>6</sub> | -3.3998                    | 0.0810                 | 0.0000        | P-1                | La       | Cr       | Ti       |
| mp-1276363  | La <sub>2</sub> TiCrO <sub>6</sub> | -3.3999                    | 0.0809                 | 0.0000        | P-1                | La       | Cr       | Ti       |
| mp-1277771  | La <sub>2</sub> TiCrO <sub>6</sub> | -3.4010                    | 0.0799                 | 0.0000        | P-1                | La       | Cr       | Ti       |
| mp-1279644  | La <sub>2</sub> TiCrO <sub>6</sub> | -3.3998                    | 0.0810                 | 0.0000        | P-1                | La       | Cr       | Ti       |
| mp-1223138  | La <sub>2</sub> TiMnO <sub>6</sub> | -3.4658                    | 0.0243                 | 2.2132        | P2 <sub>1</sub> /c | La       | Ti       | Mn       |
| mp-1079373  | La <sub>2</sub> TiNiO <sub>6</sub> | -3.2232                    | 0.1040                 | 0.0000        | Fm-3m              | La       | Ni       | Ti       |
| mp-1211325  | La <sub>2</sub> TiNiO <sub>6</sub> | -3.3161                    | 0.0110                 | 2.8034        | P2 <sub>1</sub> /c | La       | Ni       | Ti       |
| mp-1223233  | La <sub>2</sub> VFeO <sub>6</sub>  | -3.0638                    | 0.0023                 | 0.0000        | P2 <sub>1</sub> /c | La       | Fe       | V        |
| mp-1651775  | La <sub>2</sub> VFeO <sub>6</sub>  | -3.0448                    | 0.0213                 | 0.1083        | P-1                | La       | Fe       | V        |
| mp-1040836  | LaCrCoO <sub>6</sub>               | -2.0400                    | 0.1726                 | 0.9387        | Pc                 | La       | Cr       | Co       |
| mp-1040835  | LaCrReO <sub>6</sub>               | -2.6142                    | 0.1103                 | 0.0000        | P1                 | La       | Re       | Cr       |
| mp-1040819  | LaCrWO <sub>6</sub>                | -2.7214                    | 0.0828                 | 0.7814        | P1                 | La       | Cr       | W        |
| mp-1043944  | LaFeMoO <sub>6</sub>               | -2.4017                    | 0.1204                 | 1.4188        | Pc                 | La       | Fe       | Mo       |
| mp-1043955  | LaFeReO <sub>6</sub>               | -2.4271                    | 0.1065                 | 0.0000        | Pc                 | La       | Re       | Fe       |
| mp-1043919  | LaFeWO <sub>6</sub>                | -2.5326                    | 0.0800                 | 1.7784        | Pc                 | La       | Fe       | W        |
| mp-1049205  | LaMgCrMoO <sub>6</sub>             | -2.7263                    | 0.0892                 | 1.3459        | P1                 | La       | Cr       | Mo       |
| mp-2240259  | LaMgCrMoO <sub>6</sub>             | -2.6932                    | 0.1222                 | 1.3430        | P1                 | La       | Cr       | Mo       |
| mp-1044938  | LaMgCrWO <sub>6</sub>              | -2.7057                    | 0.1596                 | 0.0000        | Pc                 | La       | Cr       | W        |
| mp-2222872  | LaMgCrWO <sub>6</sub>              | -2.6927                    | 0.1726                 | 2.1621        | P1                 | La       | Cr       | W        |
| mp-1043870  | LaMgFeNiO <sub>6</sub>             | -2.1908                    | 0.1070                 | 0.0000        | Pc                 | La       | Fe       | Ni       |
| mp-1646328  | LaMgFeNiO <sub>6</sub>             | -2.1877                    | 0.1100                 | 0.0000        | P1                 | La       | Fe       | Ni       |
| mp-1043975  | LaMgFeWO <sub>6</sub>              | -2.6224                    | 0.0879                 | 2.5266        | Pc                 | La       | Fe       | W        |
| mp-1043996  | LaMgMnFeO <sub>6</sub>             | -2.5314                    | 0.1747                 | 1.4124        | Pc                 | La       | Fe       | Mn       |
| mp-1043991  | LaMgTaFeO <sub>6</sub>             | -3.0493                    | 0.0887                 | 3.1211        | Pc                 | La       | Fe       | Ta       |

| Material ID | Formula    | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|------------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-1043786  | LaMgTiFeO6 | -3.0282                    | 0.1146                 | 2.5232        | Pc          | La       | Fe       | Ti       |
| mp-1044888  | LaMgVCrO6  | -2.8906                    | 0.1205                 | 1.3575        | Pc          | La       | Cr       | V        |
| mp-1043844  | LaMgVFeO6  | -2.7014                    | 0.1474                 | 1.5532        | Pc          | La       | Fe       | V        |
| mp-1040853  | LaTaCrO6   | -3.0246                    | 0.1332                 | 0.1808        | P1          | La       | Cr       | Ta       |
| mp-1044002  | LaTaFeO6   | -2.7862                    | 0.1632                 | 0.0000        | Pc          | La       | Fe       | Ta       |
| mp-1222764  | LaTaTiO6   | -3.6440                    | 0.0099                 | 3.3911        | C2          | La       | Ti       | Ta       |
| mp-1222878  | LaTaTiO6   | -3.6539                    | 0.0000                 | 2.9188        | Pna2_1      | La       | Ti       | Ta       |
| mp-1346591  | LaTiCrO6   | -2.8513                    | 0.1713                 | 0.1022        | Pc          | La       | Cr       | Ti       |
| mp-1222787  | LaTiNbO6   | -3.5189                    | 0.0226                 | 0.0004        | C2          | La       | Nb       | Ti       |
| mp-1222933  | LaTiNbO6   | -3.5288                    | 0.0127                 | 2.7987        | Pna2_1      | La       | Nb       | Ti       |
| mp-1105823  | NaLaFeWO6  | -2.5695                    | 0.0000                 | 2.2965        | P2_1        | Na       | Fe       | W        |
| mp-1517281  | NaLaHfNbO6 | -3.2870                    | 0.1520                 | 0.0000        | F-43m       | Na       | Nb       | Hf       |
| mp-1516496  | NaLaHfTiO6 | -3.4951                    | 0.0930                 | 2.3593        | F-43m       | Na       | Hf       | Ti       |
| mp-1518151  | NaLaHfZrO6 | -3.5000                    | 0.1551                 | 2.4644        | F-43m       | Na       | Zr       | Hf       |
| mp-1517122  | NaLaMnWO6  | -2.5515                    | 0.1311                 | 1.0188        | F-43m       | Na       | Mn       | W        |
| mp-22748    | NaLaMnWO6  | -2.6827                    | 0.0000                 | 2.0322        | P2_1        | Na       | Mn       | W        |
| mp-1173651  | NaLaNiWO6  | -2.4919                    | 0.0133                 | 3.3392        | P1          | Na       | Ni       | W        |
| mp-1522096  | NaLaVWO6   | -2.6294                    | 0.1083                 | 0.4148        | F-43m       | Na       | V        | W        |
| mp-1520308  | NaLaZrFeO6 | -2.7214                    | 0.1479                 | 0.0000        | F-43m       | Na       | Fe       | Zr       |
| mp-1520149  | NaLaZrNbO6 | -3.2098                    | 0.1562                 | 0.0000        | F-43m       | Na       | Zr       | Nb       |
| mp-1521929  | NaLaZrTiO6 | -3.4046                    | 0.1138                 | 2.2686        | F-43m       | Na       | Zr       | Ti       |
| mp-1521458  | NaLaZrVO6  | -3.0644                    | 0.1685                 | 0.0000        | F-43m       | Na       | Zr       | V        |
| mp-1520467  | NaMnBiWO6  | -1.8771                    | 0.1851                 | 0.0000        | F-43m       | Na       | Mn       | W        |
| mp-1521298  | NaNbInWO6  | -2.2033                    | 0.1759                 | 0.0000        | F-43m       | Na       | Nb       | W        |
| mp-1210062  | NaNdCoWO6  | -2.4899                    | 0.0000                 | 2.4461        | P2_1        | Na       | Co       | W        |
| mp-1189143  | NaNdFeWO6  | -2.5481                    | 0.0000                 | 2.4171        | P2_1        | Na       | Fe       | W        |
| mp-1522126  | NaNdHfNbO6 | -3.2070                    | 0.1982                 | 0.0000        | F-43m       | Na       | Nb       | Hf       |

| Material ID | Formula    | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|------------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-1516737  | NaNdHfTiO6 | -3.4298                    | 0.1254                 | 2.2438        | F-43m       | Na       | Hf       | Ti       |
| mp-1517896  | NaNdMnWO6  | -2.4787                    | 0.1784                 | 1.0430        | F-43m       | Na       | Mn       | W        |
| mp-20156    | NaNdMnWO6  | -2.6572                    | 0.0000                 | 2.2832        | P2_1        | Na       | Mn       | W        |
| mp-1521141  | NaCaHfNbO6 | -3.2454                    | 0.0875                 | 2.4100        | F-43m       | Na       | Nb       | Hf       |
| mp-1516390  | NaCaHfWO6  | -2.8795                    | 0.1287                 | 0.0000        | F-43m       | Na       | Hf       | W        |
| mp-1521548  | NaCaNbWO6  | -2.6107                    | 0.1443                 | 0.0000        | F-43m       | Na       | Nb       | W        |
| mp-39712    | NaCaTaTiO6 | -3.2971                    | 0.0125                 | 2.5592        | Pc          | Na       | Ta       | Ti       |
| mp-1518125  | NaCaZrNbO6 | -3.1594                    | 0.0981                 | 2.2840        | F-43m       | Na       | Zr       | Nb       |
| mp-1516212  | NaCaZrWO6  | -2.8020                    | 0.1443                 | 0.0000        | F-43m       | Na       | Zr       | W        |
| mp-1516948  | NaEuHfFeO6 | -2.5937                    | 0.1926                 | 0.0000        | F-43m       | Na       | Fe       | Hf       |
| mp-1518553  | NaEuHfNbO6 | -3.2782                    | 0.0150                 | 0.0038        | F-43m       | Na       | Nb       | Hf       |
| mp-1517978  | NaEuHfTiO6 | -3.2848                    | 0.1473                 | 0.0000        | F-43m       | Na       | Hf       | Ti       |
| mp-1522189  | NaEuHfWO6  | -2.9099                    | 0.0855                 | 0.0000        | F-43m       | Na       | Hf       | W        |
| mp-1522034  | NaEuNbWO6  | -2.6429                    | 0.0991                 | 0.0000        | F-43m       | Na       | Nb       | W        |
| mp-1517534  | NaEuZrTiO6 | -3.1958                    | 0.1744                 | 0.0000        | F-43m       | Na       | Zr       | Ti       |
| mp-1516967  | NaEuZrWO6  | -2.8349                    | 0.0986                 | 0.0000        | F-43m       | Na       | Zr       | W        |
| mp-1220568  | Nd2CoNiO6  | -2.5203                    | 0.0000                 | 0.0000        | P2_1/c      | Nd       | Co       | Ni       |
| mp-1210035  | Nd2CoRuO6  | -2.4887                    | 0.0938                 | 0.0000        | P2_1/c      | Nd       | Co       | Ru       |
| mp-1220540  | Nd2FeNiO6  | -2.6136                    | 0.0000                 | 0.0000        | P2_1/c      | Nd       | Fe       | Ni       |
| mp-19336    | Nd2MnCoO6  | -2.7947                    | 0.0000                 | 0.0000        | P2_1/c      | Nd       | Co       | Mn       |
| mp-1220516  | Nd2MnCrO6  | -3.0322                    | 0.0205                 | 0.7902        | P2_1/c      | Nd       | Cr       | Mn       |
| mp-1220545  | Nd2MnFeO6  | -2.8589                    | 0.0111                 | 0.6503        | P2_1/c      | Nd       | Fe       | Mn       |
| mp-18899    | Nd2MnNiO6  | -2.7804                    | 0.0000                 | 1.4572        | P2_1/c      | Nd       | Mn       | Ni       |
| mp-1220508  | Nd2MnRhO6  | -2.7124                    | 0.0169                 | 0.4194        | P2_1/c      | Nd       | Mn       | Rh       |
| mp-1209990  | Nd2NiRuO6  | -2.5796                    | 0.0000                 | 0.0000        | P2_1/c      | Nd       | Ni       | Ru       |
| mp-1220528  | Nd2VFeO6   | -3.0063                    | 0.0000                 | 0.7975        | P2_1/c      | Nd       | Fe       | V        |
| mp-1518805  | NdEuHfCrO6 | -3.3186                    | 0.1265                 | 0.0215        | F-43m       | Nd       | Cr       | Hf       |

| Material ID | Formula    | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|------------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-1521819  | NdEuHfTiO6 | -3.5453                    | 0.1450                 | 0.0000        | F-43m       | Nd       | Hf       | Ti       |
| mp-1520563  | NdEuMnFeO6 | -2.5873                    | 0.1825                 | 0.0000        | F-43m       | Nd       | Fe       | Mn       |
| mp-1516183  | NdEuZrTiO6 | -3.4627                    | 0.1618                 | 0.0000        | F-43m       | Nd       | Zr       | Ti       |
| mp-1221020  | NdMnFeBiO6 | -2.1240                    | 0.1950                 | 0.0570        | P2_1        | Nd       | Fe       | Mn       |
| mp-1237102  | NdMnFeBiO6 | -2.2582                    | 0.0607                 | 0.8844        | P2_1        | Nd       | Fe       | Mn       |
| mp-1220116  | NdTaTiO6   | -3.6264                    | 0.0000                 | 3.0457        | Pna2_1      | Nd       | Ta       | Ti       |
| mp-1105199  | RbLaMnWO6  | -2.6569                    | 0.0236                 | 1.6904        | P2_1        | Rb       | Mn       | W        |
| mp-1105157  | RbNdMnWO6  | -2.6243                    | 0.0263                 | 2.6501        | P2_1        | Rb       | Mn       | W        |
| mp-1216838  | TiNbBiO6   | -2.8460                    | 0.0041                 | 2.3865        | P1          | Ti       | Nb       | Ti       |
| mp-1189894  | Y2MnCoO6   | -2.8346                    | 0.0000                 | 0.0000        | P2_1/c      | Y        | Co       | Mn       |
| mp-19192    | Y2MnNiO6   | -2.8229                    | 0.0000                 | 0.0000        | P2_1/c      | Y        | Mn       | Ni       |
| mp-1045038  | YCr3O6     | -2.6607                    | 0.1181                 | 1.6874        | P-3m1       | Y        | Cr       | Y        |
| mp-1105234  | YCrTeO6    | -2.5309                    | 0.0000                 | 2.6946        | P-31c       | Y        | Cr       | Y        |
| mp-1215956  | Yb2MnCoO6  | -2.9107                    | 0.0000                 | 0.0000        | P2_1/c      | Yb       | Co       | Mn       |
| mp-1215802  | YbTaTiO6   | -3.5128                    | 0.0000                 | 0.0000        | P2_1/c      | Yb       | Ta       | Ti       |



# Double Perovskite Curated Dataset

Supplementary Material: Curated database of double perovskite compounds and DFT-calculated properties used for training models.

## Sheet 2: Non-True Double Perovskites (N = 144 samples)

| Material ID | Formula          | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|------------------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-2230028  | Ba2La2MgMn(WO6)2 | -2.8756                    | 0.1326                 | 2.2646        | C2/m        | Ba       | Mn       | W        |
| mp-560825   | Ba2La2Mn(WO6)2   | -2.9369                    | 0.0735                 | 1.5055        | R-3m        | Ba       | Mn       | W        |
| mp-1228484  | Ba2La2Ni(WO6)2   | -2.8646                    | 0.0493                 | 3.3024        | R-3m        | Ba       | Ni       | W        |
| mp-2230422  | Ba4MgCo(ReO6)2   | -2.5826                    | 0.0840                 | 0.0000        | R-3m        | Ba       | Re       | Co       |
| mp-1228145  | Ba6Ca2Ta(RuO6)3  | -2.7241                    | 0.0000                 | 0.0000        | P3m1        | Ba       | Ta       | Ru       |
| mp-684803   | Ba3La3Mn2(WO6)3  | -2.8566                    | 0.1174                 | 1.4950        | R3m         | Ba       | Mn       | W        |
| mp-1106249  | Ba4Co(ReO6)2     | -2.6280                    | 0.0000                 | 1.5360        | R-3m        | Ba       | Re       | Co       |
| mp-1375060  | CaLa2Mn2(FeO6)2  | -2.3783                    | 0.1638                 | 0.0000        | P1          | Ca       | Fe       | Mn       |
| mp-1233435  | CaMn4(CoO6)2     | -1.7730                    | 0.0975                 | 1.1607        | C2          | Ca       | Co       | Mn       |
| mp-1234878  | CaMn4(CoO6)2     | -1.7700                    | 0.1005                 | 0.4937        | P1          | Ca       | Co       | Mn       |
| mp-1233707  | CaV4(NiO6)2      | -2.2372                    | 0.1056                 | 0.0000        | C2          | Ca       | Ni       | V        |
| mp-1233711  | CaV4(NiO6)2      | -2.1588                    | 0.1841                 | 0.9981        | P1          | Ca       | Ni       | V        |
| mp-1227837  | Ca15Ni(IrO6)4    | -2.6318                    | 0.0321                 | 0.0000        | P2          | Ca       | Ni       | Ir       |
| mp-1227895  | Ca15Ni(IrO6)4    | -2.6312                    | 0.0326                 | 0.0000        | C2          | Ca       | Ni       | Ir       |
| mp-1374973  | Ca3Co2(WO6)2     | -2.3614                    | 0.0672                 | 0.9476        | P1          | Ca       | Co       | W        |
| mp-1357678  | Ca3Cr6(FeO6)4    | -1.9989                    | 0.1437                 | 0.5526        | P-1         | Ca       | Fe       | Cr       |
| mp-1323783  | Ca3Fe2(MoO6)2    | -2.3982                    | 0.0753                 | 0.0000        | P1          | Ca       | Fe       | Mo       |
| mp-1366463  | Ca3Fe2(ReO6)2    | -2.4308                    | 0.0264                 | 1.3365        | P1          | Ca       | Re       | Fe       |
| mp-1347394  | Ca3Fe2(RhO6)2    | -1.7223                    | 0.1939                 | 0.0000        | R32         | Ca       | Fe       | Rh       |
| mp-1275790  | Ca3Fe2(WO6)2     | -2.5131                    | 0.0383                 | 0.0000        | Pm          | Ca       | Fe       | W        |
| mp-1301722  | Ca3Fe2(WO6)2     | -2.4622                    | 0.0891                 | 1.7685        | R3m         | Ca       | Fe       | W        |
| mp-1228602  | Ca3LaFe2(MoO6)2  | -2.4524                    | 0.1917                 | 0.7501        | P1          | Ca       | Fe       | Mo       |
| mp-1227846  | Ca3LaMn2(IrO6)2  | -2.4273                    | 0.0637                 | 0.0000        | P1          | Ca       | Mn       | Ir       |

| Material ID | Formula                                                            | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group          | element1 | element3 | element4 |
|-------------|--------------------------------------------------------------------|----------------------------|------------------------|---------------|----------------------|----------|----------|----------|
| mp-1340405  | Ca <sub>3</sub> Mn <sub>2</sub> (IrO <sub>6</sub> ) <sub>2</sub>   | -2.0668                    | 0.0276                 | 0.0000        | P1                   | Ca       | Mn       | Ir       |
| mp-1275109  | Ca <sub>3</sub> Mn <sub>2</sub> (WO <sub>6</sub> ) <sub>2</sub>    | -2.5897                    | 0.0424                 | 1.7350        | P1                   | Ca       | Mn       | W        |
| mp-1341945  | Ca <sub>3</sub> Ni <sub>2</sub> (WO <sub>6</sub> ) <sub>2</sub>    | -2.2566                    | 0.1056                 | 0.0000        | P1                   | Ca       | Ni       | W        |
| mp-1043066  | Ca <sub>3</sub> Ta <sub>6</sub> (FeO <sub>6</sub> ) <sub>4</sub>   | -2.8836                    | 0.1177                 | 2.7703        | P-1                  | Ca       | Fe       | Ta       |
| mp-1214146  | Ca <sub>3</sub> V <sub>3</sub> (CoO <sub>6</sub> ) <sub>2</sub>    | -2.5376                    | 0.0577                 | 0.0000        | Ia-3d                | Ca       | Co       | V        |
| mp-1214162  | Ca <sub>3</sub> V <sub>3</sub> (NiO <sub>6</sub> ) <sub>2</sub>    | -2.5660                    | 0.0165                 | 0.0000        | Ia-3d                | Ca       | Ni       | V        |
| mp-1042908  | Ca <sub>3</sub> V <sub>6</sub> (FeO <sub>6</sub> ) <sub>4</sub>    | -2.4155                    | 0.0160                 | 1.7641        | P-1                  | Ca       | Fe       | V        |
| mp-1329093  | Ca <sub>5</sub> Mn <sub>2</sub> (NiO <sub>6</sub> ) <sub>2</sub>   | -2.2871                    | 0.1040                 | 0.0000        | C2                   | Ca       | Mn       | Ni       |
| mp-1227125  | Ca <sub>7</sub> Ni(IrO <sub>6</sub> ) <sub>2</sub>                 | -2.5278                    | 0.0411                 | 0.0866        | R32                  | Ca       | Ni       | Ir       |
| mp-1227618  | Ca <sub>9</sub> Y <sub>3</sub> (CoO <sub>6</sub> ) <sub>4</sub>    | -2.5356                    | 0.0535                 | 0.0000        | P1                   | Ca       | Co       | Y        |
| mp-1227804  | Ca <sub>9</sub> Y <sub>3</sub> (CoO <sub>6</sub> ) <sub>4</sub>    | -2.5384                    | 0.0507                 | 0.0000        | P1                   | Ca       | Co       | Y        |
| mp-1225913  | Eu <sub>2</sub> Mo(WO <sub>6</sub> ) <sub>2</sub>                  | -2.4491                    | 0.0490                 | 0.0000        | C2/c                 | Eu       | Mo       | W        |
| mp-560123   | FeBi <sub>3</sub> (MoO <sub>6</sub> ) <sub>2</sub>                 | -1.8474                    | 0.0025                 | 1.2333        | C2/c                 | Fe       | Fe       | Mo       |
| mp-2231116  | La <sub>2</sub> MgCr <sub>2</sub> (CoO <sub>6</sub> ) <sub>2</sub> | -2.2545                    | 0.1832                 | 0.4194        | P1                   | La       | Cr       | Co       |
| mp-2230275  | La <sub>2</sub> MgFe <sub>2</sub> (MoO <sub>6</sub> ) <sub>2</sub> | -2.4862                    | 0.1358                 | 1.0145        | P1                   | La       | Fe       | Mo       |
| mp-2230665  | La <sub>2</sub> MgFe <sub>2</sub> (WO <sub>6</sub> ) <sub>2</sub>  | -2.5734                    | 0.1140                 | 1.4957        | P1                   | La       | Fe       | W        |
| mp-2231608  | La <sub>2</sub> MgFe <sub>2</sub> (WO <sub>6</sub> ) <sub>2</sub>  | -2.5648                    | 0.1226                 | 1.0791        | P1                   | La       | Fe       | W        |
| mp-1220995  | NaNdTiNbO <sub>6</sub> F                                           | -3.3645                    | 0.0432                 | 2.7683        | Imma                 | Na       | Nb       | Ti       |
| mp-43134    | NaNdTiNbO <sub>6</sub> F                                           | -3.3685                    | 0.0392                 | 2.9335        | P4 <sub>1</sub> 22   | Na       | Nb       | Ti       |
| mp-694918   | NaCa <sub>2</sub> V <sub>3</sub> (CoO <sub>6</sub> ) <sub>2</sub>  | -2.4751                    | 0.0000                 | 2.1268        | I4 <sub>1</sub> /acd | Na       | V        | Co       |
| mp-1221057  | NaEuTiNbO <sub>6</sub> F                                           | -3.2126                    | 0.0331                 | 0.0000        | Imma                 | Na       | Nb       | Ti       |
| mp-43048    | NaEuTiNbO <sub>6</sub> F                                           | -3.2160                    | 0.0298                 | 0.0000        | P4 <sub>1</sub> 322  | Na       | Nb       | Ti       |
| mp-1221023  | NaGdTINbO <sub>6</sub> F                                           | -3.3611                    | 0.0529                 | 0.0000        | Imma                 | Na       | Nb       | Ti       |
| mp-42981    | NaGdTINbO <sub>6</sub> F                                           | -3.3665                    | 0.0475                 | 2.4183        | P4 <sub>1</sub> 322  | Na       | Nb       | Ti       |
| mp-1219608  | RbLaTiNbO <sub>6</sub> F                                           | -3.3032                    | 0.1294                 | 1.0761        | Pmm2                 | Rb       | Nb       | Ti       |
| mp-1219593  | RbNdTiNbO <sub>6</sub> F                                           | -3.3645                    | 0.0436                 | 1.8523        | P4mm                 | Rb       | Nb       | Ti       |
| mp-1324685  | Y <sub>5</sub> Mn <sub>3</sub> (NiO <sub>6</sub> ) <sub>3</sub>    | -2.4931                    | 0.0820                 | 0.0000        | Pc                   | Y        | Mn       | Ni       |

| Material ID | Formula           | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|-------------------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-686765   | YTi2CdO6F         | -3.1777                    | 0.1427                 | 1.3491        | Imma        | Y        | Cd       | Ti       |
| mp-686330   | Ba10Mg3ZrTa6O30   | -3.3241                    | 0.1116                 | 0.0000        | R-3m        | Ba       | Zr       | Ta       |
| mp-1228979  | Ba10Ti8NbNiO30    | -3.3239                    | 0.0128                 | 0.0000        | I4/mmm      | Ba       | Nb       | Ni       |
| mp-720103   | Ba4La6Mn(Co3O10)3 | -2.3389                    | 0.1650                 | 0.0000        | P4/mmm      | Ba       | Mn       | Co       |
| mp-1228585  | Ba4Ta10CoO30      | -3.3383                    | 0.0000                 | 2.4312        | Imm2        | Ba       | Co       | Ta       |
| mp-1228547  | Ba4Ta10FeO30      | -3.3527                    | 0.0000                 | 2.0407        | Imm2        | Ba       | Fe       | Ta       |
| mp-1228660  | Ba4Ta10NiO30      | -3.3224                    | 0.0073                 | 0.0000        | Imm2        | Ba       | Ta       | Ni       |
| mp-1228527  | Ba6Nb9CoO30       | -3.0755                    | 0.0477                 | 0.0000        | Cmm2        | Ba       | Co       | Nb       |
| mp-695409   | Ba7Na3Ti7Nb3O30   | -3.2708                    | 0.0273                 | 1.1140        | Pm          | Ba       | Nb       | Ti       |
| mp-1228755  | Ba8ZrNb19O36      | -2.9103                    | 0.0000                 | 0.0000        | P4/mmm      | Ba       | Nb       | Zr       |
| mp-557475   | BaBi12Mo4WO34     | -1.9166                    | 0.0183                 | 3.3477        | P2/c        | Ba       | Mo       | W        |
| mp-1214892  | BaCr2(RuO3)4      | -1.8522                    | 0.0375                 | 0.0000        | P4          | Ba       | Cr       | Ru       |
| mp-1021950  | BaNb2(FeO3)4      | -2.3135                    | 0.1083                 | 1.7345        | I4/m        | Ba       | Fe       | Nb       |
| mp-19502    | BaTi18Mn3O38      | -3.3327                    | 0.0160                 | 0.0000        | R-3         | Ba       | Mn       | Ti       |
| mp-1048316  | Ba2YCo3O7         | -2.1806                    | 0.1561                 | 0.0000        | Pmmm        | Ba       | Co       | Y        |
| mp-1048020  | Ba2YCo3O8         | -2.2305                    | 0.0636                 | 0.0000        | P4/mmm      | Ba       | Co       | Y        |
| mp-2228739  | Ba2YMgCo3O8       | -2.2507                    | 0.1796                 | 0.0000        | Pmm2        | Ba       | Co       | Y        |
| mp-704461   | Ba3La7Mn(Co3O10)3 | -2.4114                    | 0.1477                 | 0.0000        | P2/m        | Ba       | Mn       | Co       |
| mp-697840   | Ba3La7Ti3Cr7O30   | -3.2558                    | 0.0295                 | 1.8541        | P1          | Ba       | Cr       | Ti       |
| mp-676975   | Ba3Na7Ti3Nb7O30   | -2.9995                    | 0.0396                 | 1.1087        | Pm          | Ba       | Nb       | Ti       |
| mp-694926   | CaLa9Ti5Cr5O30    | -3.3985                    | 0.0642                 | 0.0000        | P1          | Ca       | Cr       | Ti       |
| mp-743819   | Ca10Ti8NbFeO30    | -3.4292                    | 0.0040                 | 1.9961        | P-1         | Ca       | Fe       | Nb       |
| mp-743953   | Ca2La8Ti5Cr5O30   | -3.3969                    | 0.0475                 | 0.0000        | P-1         | Ca       | Cr       | Ti       |
| mp-744126   | Ca3La7Ti5Cr5O30   | -3.3802                    | 0.0452                 | 0.0000        | Pm          | Ca       | Cr       | Ti       |
| mp-695081   | Ca5Ho3Ti5Mn3O24   | -3.3507                    | 0.0226                 | 0.5773        | P1          | Ca       | Mn       | Ti       |
| mp-743863   | Ca5La7Mn7Ru5O36   | -2.6645                    | 0.1292                 | 0.0000        | P1          | Ca       | Mn       | Ru       |
| mp-699041   | Ca5Nd5Mn9CrO30    | -2.7641                    | 0.0060                 | 0.0000        | P1          | Ca       | Cr       | Mn       |

| Material ID | Formula         | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|-----------------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-1227118  | Ca6Co3RhO12     | -2.3594                    | 0.0318                 | 0.1344        | R-3         | Ca       | Co       | Rh       |
| mp-695477   | Ca6La4Ti5Cr5O30 | -3.3043                    | 0.0262                 | 0.1654        | P1          | Ca       | Cr       | Ti       |
| mp-532691   | Ca6Nd6Mn11CrO36 | -2.7426                    | 0.0246                 | 0.0000        | P1          | Ca       | Cr       | Mn       |
| mp-744008   | Ca7La3Ti5Cr5O30 | -3.2570                    | 0.0207                 | 0.0000        | Pm          | Ca       | Cr       | Ti       |
| mp-695500   | Ca8La2Ti5Cr5O30 | -3.2024                    | 0.0227                 | 0.0000        | P-1         | Ca       | Cr       | Ti       |
| mp-744380   | Ca9LaTi4Mn6O30  | -2.9816                    | 0.0310                 | 0.0000        | P1          | Ca       | Mn       | Ti       |
| mp-706246   | Ca9LaTi6Mn4O30  | -3.1839                    | 0.0329                 | 0.0000        | P1          | Ca       | Mn       | Ti       |
| mp-694916   | Ca9LaTi8Mn2O30  | -3.3913                    | 0.0295                 | 0.0000        | P1          | Ca       | Mn       | Ti       |
| mp-694954   | Ca9LaTi9MnO30   | -3.4933                    | 0.0215                 | 0.0000        | Pm          | Ca       | Mn       | Ti       |
| mp-1214052  | CaFe(MoO3)2     | -2.0306                    | 0.1818                 | 0.0000        | P2_1/c      | Ca       | Fe       | Mo       |
| mp-551086   | CrFe(BiO3)2     | -1.8234                    | 0.0159                 | 0.0000        | R3          | Cr       | Cr       | Fe       |
| mp-1181596  | Fe3Co(BiO3)4    | -1.5394                    | 0.0812                 | 0.7248        | P1          | Fe       | Co       | Fe       |
| mp-1225360  | Fe4Co(BiO3)5    | -1.5904                    | 0.0409                 | 1.0820        | P1          | Fe       | Co       | Fe       |
| mp-1225635  | Fe4Co(BiO3)5    | -1.5859                    | 0.0454                 | 1.4885        | P1          | Fe       | Co       | Fe       |
| mp-1225879  | Fe4Co(BiO3)5    | -1.5852                    | 0.0461                 | 1.3264        | P1          | Fe       | Co       | Fe       |
| mp-1225028  | FeCo(BiO3)2     | -1.5189                    | 0.0478                 | 0.5355        | P4mm        | Fe       | Co       | Fe       |
| mp-1223593  | KBa3Y2(Co3O8)2  | -2.0396                    | 0.1611                 | 0.0000        | I4mm        | K        | Co       | Y        |
| mp-743640   | Li3La8Ti9Cr3O36 | -3.4242                    | 0.0669                 | 1.4458        | P1          | Li       | Cr       | Ti       |
| mp-1307408  | Li3Y(NiO3)2     | -2.0049                    | 0.0763                 | 0.1806        | P-1         | Li       | Ni       | Y        |
| mp-1309972  | Li3Y(NiO3)2     | -2.0152                    | 0.0660                 | 0.7283        | C2          | Li       | Ni       | Y        |
| mp-754436   | Li3Y(NiO3)2     | -2.0050                    | 0.0762                 | 0.0000        | C2/m        | Li       | Ni       | Y        |
| mp-1221980  | Mn2Ni(BiO3)3    | -1.6433                    | 0.0719                 | 0.0000        | C2/c        | Mn       | Mn       | Ni       |
| mp-545769   | MnCo(BiO3)2     | -1.6311                    | 0.0286                 | 0.0000        | C2          | Mn       | Co       | Mn       |
| mp-1221875  | MnFe3(BiO3)4    | -1.6766                    | 0.0394                 | 0.9373        | P1          | Mn       | Fe       | Mn       |
| mp-1222178  | MnFe4(BiO3)5    | -1.6682                    | 0.0395                 | 1.0898        | P1          | Mn       | Fe       | Mn       |
| mp-1222245  | MnFe4(BiO3)5    | -1.6541                    | 0.0535                 | 0.0000        | P1          | Mn       | Fe       | Mn       |
| mp-23618    | MnNi(BiO3)2     | -1.6133                    | 0.0229                 | 0.8542        | C2/c        | Mn       | Mn       | Ni       |

| Material ID | Formula            | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|--------------------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-555839   | MnNi(BiO3)2        | -1.6210                    | 0.0152                 | 0.0000        | P2_1/c      | Mn       | Mn       | Ni       |
| mp-562612   | MnNi(BiO3)2        | -1.6139                    | 0.0224                 | 0.0000        | C2          | Mn       | Mn       | Ni       |
| mp-694044   | Na3Ca7Ta3Ti7O30    | -3.3979                    | 0.0103                 | 2.3925        | P1          | Na       | Ta       | Ti       |
| mp-695506   | Na3Ca7Ti7Nb3O30    | -3.2767                    | 0.0660                 | 1.1805        | Pm          | Na       | Nb       | Ti       |
| mp-1173864  | Na3La8Ti9Mn3O36    | -3.3715                    | 0.0714                 | 1.2517        | P1          | Na       | Mn       | Ti       |
| mp-1173810  | Na2La7Ti8Mn2O30    | -3.4587                    | 0.0529                 | 0.2925        | P1          | Na       | Mn       | Ti       |
| mp-721094   | Na5Ca2Ce3Ti8Nb2O30 | -3.2865                    | 0.0407                 | 0.0000        | P4mm        | Na       | Nb       | Ti       |
| mp-693144   | Na9CaTa9TiO30      | -3.0182                    | 0.0941                 | 2.5387        | Pm          | Na       | Ta       | Ti       |
| mp-677027   | NaCa9TaTi9O30      | -3.5050                    | 0.0018                 | 2.4407        | Pm          | Na       | Ta       | Ti       |
| mp-1220715  | Nb3Fe(BiO3)6       | -2.2126                    | 0.0003                 | 2.3257        | P1          | Nb       | Fe       | Nb       |
| mp-695426   | Na7Ca3Ta7Ti3O30    | -3.1988                    | 0.0122                 | 2.3227        | P1          | Na       | Ta       | Ti       |
| mp-1217245  | Ti3Fe(BiO3)5       | -2.3307                    | 0.0428                 | 1.6518        | Cm          | Ti       | Fe       | Ti       |
| mp-1217452  | Ti3Fe(BiO3)5       | -2.3600                    | 0.0134                 | 2.0168        | P2_1        | Ti       | Fe       | Ti       |
| mp-1217427  | Ti3Fe2(BiO3)6      | -2.2040                    | 0.0531                 | 1.5770        | Cm          | Ti       | Fe       | Ti       |
| mp-1217638  | Ti3Fe3(BiO3)7      | -2.1139                    | 0.0575                 | 1.6492        | Cm          | Ti       | Fe       | Ti       |
| mp-1208356  | Ti4Fe(BiO3)6       | -2.3767                    | 0.0567                 | 0.4693        | Fmm2        | Ti       | Fe       | Ti       |
| mp-1217566  | Ti4Nb(BiO3)7       | -2.4731                    | 0.0290                 | 2.1789        | C2          | Ti       | Nb       | Ti       |
| mp-550745   | TiFe(BiO3)2        | -2.1782                    | 0.0154                 | 0.0000        | P1          | Ti       | Fe       | Ti       |
| mp-1216832  | TiNb(BiO3)3        | -2.3672                    | 0.0685                 | 1.7575        | I4mm        | Ti       | Nb       | Ti       |
| mp-1216863  | TiNb(BiO3)3        | -2.4319                    | 0.0038                 | 2.4233        | P2_1        | Ti       | Nb       | Ti       |
| mp-1273178  | YCoO3              | -2.5803                    | 0.0117                 | 0.3490        | P-1         | Y        | Co       | Y        |
| mp-1288734  | YCoO3              | -2.5764                    | 0.0155                 | 0.1141        | P2_1/c      | Y        | Co       | Y        |
| mp-1330514  | YCoO3              | -2.5692                    | 0.0227                 | 0.0073        | P6_3cm      | Y        | Co       | Y        |
| mp-1387356  | YCoO3              | -2.4856                    | 0.1063                 | 0.0000        | P6_3/mmc    | Y        | Co       | Y        |
| mp-22668    | YCoO3              | -2.5912                    | 0.0007                 | 0.0000        | Pnma        | Y        | Co       | Y        |
| mp-1232521  | YCr(BO3)2          | -3.0625                    | 0.0087                 | 3.2276        | R-3         | Y        | Cr       | Y        |
| mp-1336975  | YCrO3              | -3.0331                    | 0.1579                 | 1.3989        | P6_3cm      | Y        | Cr       | Y        |

| Material ID | Formula    | Formation Energy (eV/atom) | Energy Above Hull (eV) | Band Gap (eV) | Space Group | element1 | element3 | element4 |
|-------------|------------|----------------------------|------------------------|---------------|-------------|----------|----------|----------|
| mp-18725    | YCrO3      | -3.1910                    | 0.0000                 | 2.4831        | Pnma        | Y        | Cr       | Y        |
| mp-1037203  | YHfMg30O32 | -2.9421                    | 0.1078                 | 0.0000        | P4/mmm      | Y        | Hf       | Y        |
| mp-1037237  | YHfMg30O32 | -2.9399                    | 0.1100                 | 0.0000        | P4/mmm      | Y        | Hf       | Y        |
| mp-1037428  | YMg30CdO32 | -2.9456                    | 0.0783                 | 0.6155        | P4/mmm      | Y        | Cd       | Y        |
| mp-1037436  | YMg30CdO32 | -2.9418                    | 0.0821                 | 0.0000        | P4/mmm      | Y        | Cd       | Y        |
| mp-1037750  | YMg30CoO32 | -2.9739                    | 0.0488                 | 1.0697        | P4/mmm      | Y        | Co       | Y        |
| mp-1037752  | YMg30CoO32 | -2.9721                    | 0.0506                 | 0.9784        | P4/mmm      | Y        | Co       | Y        |
| mp-1037254  | YMg30CrO32 | -2.9647                    | 0.0702                 | 0.0000        | P4/mmm      | Y        | Cr       | Y        |
| mp-1037480  | YMg30CrO32 | -2.9659                    | 0.0691                 | 0.0000        | P4/mmm      | Y        | Cr       | Y        |
| mp-1037756  | YMg30NiO32 | -2.9629                    | 0.0597                 | 2.0776        | P4/mmm      | Y        | Ni       | Y        |
| mp-1037774  | YMg30NiO32 | -2.9656                    | 0.0570                 | 1.1713        | P4/mmm      | Y        | Ni       | Y        |
| mp-1346989  | YNiO3      | -2.3953                    | 0.0704                 | 0.0065        | P6_3cm      | Y        | Ni       | Y        |
| mp-19242    | YNiO3      | -2.4657                    | 0.0000                 | 0.0000        | Pnma        | Y        | Ni       | Y        |